

Week 9: Lab

INFT 1103: Operating System Security

Durham College

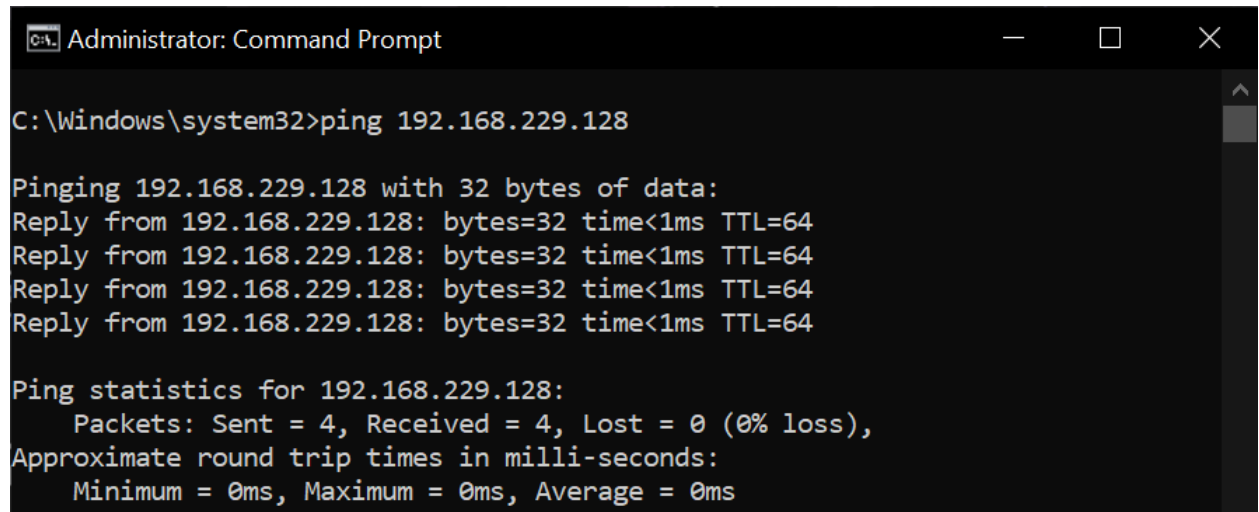
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November 11, 2025

1. Use the command *ip a* to confirm your server's IP address
 - a. Take a screenshot of your Ubuntu server's IP address and paste it here

```
melvin@ubuntu:~$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:6a:27:46 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.229.128/24 metric 100 brd 192.168.229.255 scope global dynamic ens33
        valid_lft 1789sec preferred_lft 1789sec
    inet6 fe80::20c:29ff:fe6a:2746/64 scope link
        valid_lft forever preferred_lft forever
melvin@ubuntu:~$ _
```

2. Ping the IP from your Windows host computer
 - a. Take a screenshot of the successful pings and paste it here



The screenshot shows a Windows Command Prompt window titled "Administrator: Command Prompt". The command entered is "C:\Windows\system32>ping 192.168.229.128". The output shows four successful replies from 192.168.229.128, each with 32 bytes of data, a time less than 1ms, and a TTL of 64. Below the replies, the ping statistics are displayed: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), and Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms.

```
Administrator: Command Prompt

C:\Windows\system32>ping 192.168.229.128

Pinging 192.168.229.128 with 32 bytes of data:
Reply from 192.168.229.128: bytes=32 time<1ms TTL=64
Reply from 192.168.229.128: bytes=32 time<1ms TTL=64
Reply from 192.168.229.128: bytes=32 time<1ms TTL=64
Reply from 192.168.229.128: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.229.128:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

3. On your local Windows computer (NOT your ubuntu machine) create an ssh keypair: open command prompt and type: `ssh-keygen`
 - a. Take a screenshot of the confirmation message, including the fingerprint, and paste it [here](#)

```
C:\Windows\system32>ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\Mel\.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\Mel\.ssh/id_ed25519
Your public key has been saved in C:\Users\Mel\.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:sYnRKRGifdj1dhDXyIgDDSMJj/z1Bqtli+MfT6pHQE mel@DESKTOP-F3NGS7H
The key's randomart image is:
+--[ED25519 256]--+
|    =+==..*o . .  |
| . + =o.+o.+o o   |
| + .E+o.+ . .     |
| + o.*+ +         |
| = *o.S           |
| * .              |
| = ...            |
| . +.=.           |
| . =o.            |
+-----[SHA256]-----+
C:\Windows\system32>
```

4. Once the keypair is created, copy the public key to a safe directory with the command: `copy C:\Users\yourUserName\.ssh\id_rsa.pub C:\Users\yourUserName\authorized_keys`
5. Go back to your Ubuntu server
6. Start the SSH service using the commands:
 - a. `sudo apt install openssh-server`
 - b. `sudo ufw allow ssh`
7. Create an ssh directory with the command: `mkdir ~/.ssh/`
8. Go back to your Windows host computer
9. Open command prompt and use the command: `cd C:\Users\yourUserName\`
10. Copy your public ssh key to the ubuntu server using the command: `scp authorized_keys username@UbutnuServerIP:~/.ssh`

11. Go back to your Ubuntu server
12. Adjust the permission settings for the ssh share and public ssh key file:
 - a. `chmod 700 ~/.ssh`
 - b. `chmod 600 ~/.ssh/authorized_keys`
 - c. Run the `-ls -l` command on the `/.ssh` directory to ensure the changes were made. *Take a screenshot and paste it here*

```
melvin@ubuntu:~/.ssh$ ls -l
total 4
-rw----- 1 melvin melvin 411 Nov 11 14:37 authorized_keys
melvin@ubuntu:~/.ssh$ _
```

13. Go back to your Windows host computer
14. Log into your Ubuntu machine from the Windows command prompt with the command: `ssh username@UbuntuServerIP`
 - a. You should not be prompted for a password
 - b. Take a screenshot showing both the ssh login command and the ubuntu terminal (expand the command prompt window if necessary)

```
C:\Users\krist>ssh administrator@192.168.50.9
*****
This server is used for INFT1103 for demo
purposes only.

UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED

You must have explicit, authorized permission to
access or configure this device. Unauthorized
attempts and actions to access or use this system
may result in civil and/or criminal penalties.
All activities performed on this device are
logged and monitored.

*****

Welcome to Ubuntu 22.04.1 LTS (GNU/Linux 5.15.0-43-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

System information as of Tue Oct 18 02:41:09 PM UTC 2022

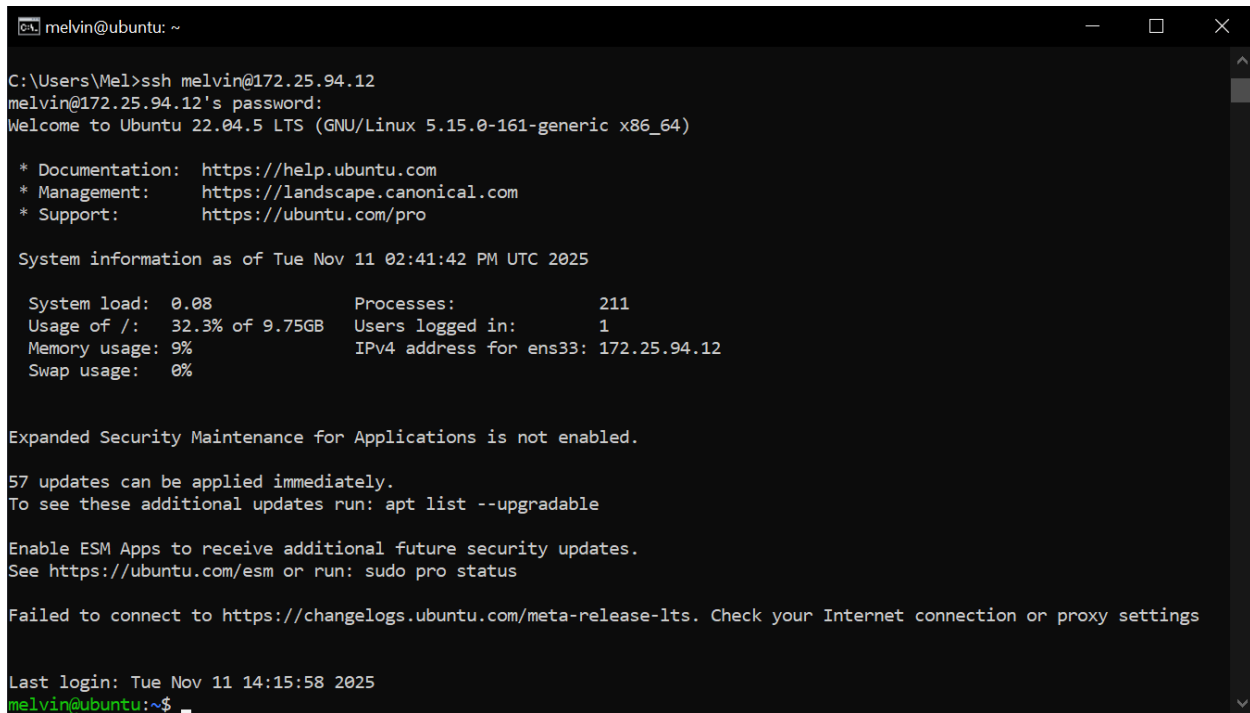
System load:  0.0               Processes:    215
Usage of /:   39.1% of 18.53GB   Users logged in: 1
Memory usage: 42%              IPv4 address for ens33: 192.168.50.9
Swap usage:   0%

0 updates can be applied immediately.

Last login: Sun Oct 16 21:53:44 2022 from 192.168.50.171
administrator@ubuntuserver:~$
```

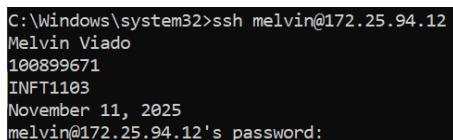
i. Example:

ii. Paste your screenshot here



```
melvin@ubuntu: ~  
C:\Users\Mel>ssh melvin@172.25.94.12  
melvin@172.25.94.12's password:  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-161-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Tue Nov 11 02:41:42 PM UTC 2025  
  
System load:  0.08          Processes:      211  
Usage of /:   32.3% of 9.75GB Users logged in: 1  
Memory usage: 9%          IPv4 address for ens33: 172.25.94.12  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
57 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings  
  
Last login: Tue Nov 11 14:15:58 2025  
melvin@ubuntu:~$
```

15. Create a banner message for your SSH service
 - a. `sudo nano /etc/ssh/ssh_banner.net`
 - b. In the text editor enter the following information:
 - i. Your Name:
 - ii. Your student number:
 - iii. The course number:
 - iv. Today's date
 - c. Close the text editor with Ctrl +X
16. Add the banner message to the SSH config file
 - a. `sudo nano /etc/ssh/sshd_config`
 - b. Find the line that says "Banner" and add the following information:
 - i. `Banner /etc/ssh/ssh_banner.net`
 - c. Close the text editor using CTRL + X
 - d. Restart the SSH service
 - i. `sudo systemctl restart ssh`
17. Log out of the SSH terminal session completely
18. Log back in using SSH
 - a. Take a screenshot of your banner message and paste it here.



```
C:\Windows\system32>ssh melvin@172.25.94.12  
Melvin Viado  
100899671  
INFT1103  
November 11, 2025  
melvin@172.25.94.12's password:
```

19. Enable ufw

a. *sudo ufw enable*

20. Enable rate limiting for port 22

a. *sudo ufw limit 22*

b. confirm the rule is enabled using *sudo ufw status*. Take a screenshot and paste it here.

```
melvin@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
22 LIMIT Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
22 (v6) LIMIT Anywhere (v6)

melvin@ubuntu:~$
```

21. Adjust the firewall's logging from low to medium

a. *ufw logging medium*

22. Log out of your SSH session and log back in

23. Read the firewall's log messages

a. *Sudo cat /var/log/ufw.log*

b. Take a screenshot of the log messages and paste it here.

```
melvin@ubuntu:~$ sudo cat /var/log/ufw.log
[sudo] password for melvin:
sudo: cat /var/log/ufw.log: command not found
melvin@ubuntu:~$ sudo cat /var/log/ufw.log
Nov 11 14:52:58 ubuntu kernel: [ 2242.102649] [UFW AUDIT] IN=ens33 OUT= MAC=00:0c:29:6a:27:46:60:dd:8e:e3:99:1e:08:00 SRC=172.25.94.101 DST=172.25.94.12 LEN=52 TOS=0x00 PREC=0x00 TTL=128 ID=26481 DF PROTO=TCP SPT=6160 DPT=22 WINDOW=64240 RES=0x00 SYN URG=0
melvin@ubuntu:~$
```

24. Install the ClamAV antivirus

a. *apt-get install clamav clamav-daemon*

25. Update your ClamAV database

a. *sudo systemctl stop clamav-freshclam*

b. *sudo freshclam*

c. *systemctl start clamav-freshclam*

d. *systemctl enable clamav-freshclam*

26. Perform an infected scan on your own user's home directory

a. *clamscan infected /home/username*

b. Take a screenshot of the results and paste it here

```
melvin@ubuntu:~$ clamscan infected /home/melvin
Loading: 19s, ETA: 0s [=====] 8.71M/8.71M sigs
Compiling: 3s, ETA: 0s [=====] 41/41 tasks

infected: No such file or directory
WARNING: infected: Can't access file
/home/melvin/.profile: OK
/home/melvin/.bashrc: OK
/home/melvin/.bash_logout: OK
/home/melvin/.sudo_as_admin_successful: Empty file
/home/melvin/.bash_history: OK

----- SCAN SUMMARY -----
Known viruses: 8708724
Engine version: 1.4.3
Scanned directories: 1
Scanned files: 4
Infected files: 0
Data scanned: 0.00 MB
Data read: 0.00 MB (ratio 0.00:1)
Time: 22.427 sec (0 m 22 s)
Start Date: 2025:11:11 14:56:14
End Date: 2025:11:11 14:56:36
melvin@ubuntu:~$
```